# Anish Kanade

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## SUMMARY

**Full-stack Software Engineer** with hands-on experience building production web, cloud, data, and machine learning systems using **Python, TypeScript/Node.js, Java, C++, SQL, and AWS**. Skilled in REST APIs, OOP, testing, relational databases, CI/CD, debugging, and Agile development. Promoted AI Data Generalist with additional hands-on experience in robotics data collection, QA, troubleshooting, and evaluation workflows. Published researcher with an **ACM SIGGRAPH 2024** presentation and publication, bringing strong problem-solving and cross-functional collaboration to software development. Hands-on experience in robotics, AI data workflows, and evaluation.

## WORK EXPERIENCE

**AI Data Generalist II** August 2026 - Present

Generalist AI, Inc.

- Expanded responsibility from data preprocessing and QA into **end-to-end robotic data collection and evaluation**, operating robots during training and evaluation runs, monitoring task performance, and producing reliable data for model development.
- **Performed human and robot QA, corrected annotations and robot behavior, and troubleshot recurring data and software issues**, identifying failures that could affect model performance and downstream training.
- **Owned task setup, resets, behavior correction, dataset validation, and versioning workflows**, translating evaluation observations and operational feedback into actionable improvements for AI and model training workflows.
- **Executed hands-on robot operations and assembly of robot hands and stands using precise, repeatable procedures**, collaborating with teammates to maintain consistent and safe system setups.

**AI Data Generalist I** March 2026 - August 2026

Generalist AI, Inc.

- Built and maintained **data pipelines** for embodied AI and robotics workflows, transforming raw sensor and operational data into clean, versioned datasets used directly in model training and evaluation loops.
- Designed and enforced **data validation** and **QA checks** across annotation pipelines, catching labeling inconsistencies early and improving dataset reliability for retraining cycles.
- Owned **preprocessing** and **dataset versioning** workflows for robot learning experiments, enabling reproducible training runs and faster iteration between research and engineering teams.
- Partnered with **engineers and researcher**s to diagnose data quality issues, refine collection procedures, and translate operational feedback into concrete **pipeline improvements**.

**Software Engineer (AI/ML)** April 2024 - March 2026

University of Massachusetts Boston

- Built and maintained a **Django-based production ML platform** that analyzes student facial expressions to help professors determine whether students have understood the topic, **enhancing classroom insights** and increasing instructor adoption by **enabling data-driven teaching adjustments.**
- **Led** data labeling and QA effort, and implemented an **end-to-end preprocessing** and annotation **pipeline on Amazon Mechanical Turk**, supporting over 5,000 facial images with **S3-backed raw/labeled storage**, built-in quality checks, **improving labeled coverage and training robustness**.
- Developed and trained **emotion-recognition models** combining CNNs and LSTMs with attention (**TensorFlow/Keras**), using versioned, reproducible training datasets to produce reliable sequence-level predictions for research experiments.
- Implemented reusable AU extraction workflows with **data validation checks**, source-to-target mappings, and documented operational procedures to ensure consistent, **analysis-ready datasets** and **reliable retraining cycles.**

## ACHIEVEMENTS

- **ACM SIGGRAPH 2024 – Conference Presentation:** Co-presented "*An Iterative Approach to Build a Semantic Dataset for Facial Expression of Personality,*" showcasing methodology advancements in HCI.
- **ACM SIGGRAPH 2024 – Publication:** "*An Iterative Approach to Build a Semantic Dataset for Facial Expression of Personality.*"

## PROJECTS

**Secure Authentication & User Management API (**Node.js, TypeScript, Fastify, Prisma, SQL, JWT, bcrypt, Jest, Supertest, React**)**

- Built a **production Fastify REST API** with secure signup/login, role-based access control, and structured data ingestion for downstream analytics.
- Designed a normalized Prisma/SQL schema with indexed constraints, JSON-schema validation, and **secure authentication** to ensure data integrity.
- Improved production reliability with **integration testing**, concurrency fixes, and documented operational workflows to support stable deployments.

**Recommendation Platform for Hikers: "HikeMates" (**React, CSS, JavaScript, HTML, Git, Agile **)**

- Architected responsive web components, **reducing layout-related issues** through refactors and **DRY principles.**
- Refactored **20+ reusable components** using **CSS Modules** to improve maintainability and speed feature development.
- **Led Agile sprint planning**, coordinating with backend engineers and product stakeholders to define APIs and deliver MVP features on schedule.

**Cloud-Powered Stock Analysis (**Python, AWS, Pandas, Matplotlib, Twelve Data API**)**

- Designed and automated a data pipeline to retrieve, process, and **visualize five years of stock data with reliable accuracy and consistency.**
- Used **AWS S3 & EC2** for seamless cloud integration and real-time visualization updates, **reducing latency and improving analyst workflows.**
- Collaborated with stakeholders to prioritize metrics, delivering actionable visualizations for investment simulations.

**Bank Churning Analysis (**Python, Pandas, Matplotlib, Scikit-learn**)**

- Conducted an in-depth customer churn analysis to identify attrition drivers and translate findings into clear stakeholder recommendations.
- Built a Logistic Regression model in Scikit-learn that achieved 81% test accuracy; validated via 5-fold cross-validation.
- Developed predictive churn models and recommended prioritized interventions that were projected to reduce churn and guide retention planning.

**Markov Text Generator: Probabilistic Sequence Modeling (**Python, NumPy**)**

- Built a probabilistic text generation model in Python/NumPy using Markov chains with **OOP** design principles.
- Optimized matrix operations for scalability, laying the groundwork for training on larger language corpora.

## EDUCATION

**Bachelor of Science in Computer Science** May 2025

University of Massachusetts Boston

## SKILLS

**Languages and Frameworks:** Java, Python, C, C++, SQL, MySQL (AWS RDS), TypeScript, JavaScript (HTML, CSS, React, Node.js, **Fastify**)
**Machine Learning & Models:** TensorFlow/Keras, Scikit-learn, ETL/ELT, CNNs, LSTMs, Evaluation/Deployment, Data Labeling (Amazon Mechanical Turk)
**Cloud and DevOps:** Google Cloud Platform (GCP), **AWS (S3, EC2, Lambda**), **Docker**, **RESTful API development**, **GitHub Actions (CI/CD)**
**AI Agent & Agent Workflows:** Git, GitHub Copilot (Cloud), Kiro (Agentic AI IDE), GitHub Spec Kit (Spec-driven dev), Hugging Face, LangChain